

1. Explain Threat Intelligence

Threat Intelligence is the process of collecting, analyzing, and sharing information about current and potential cyber threats. It helps organizations understand who the attackers are, how they operate, what their objectives are, and how to defend against them.

Instead of reacting only after an attack occurs, threat intelligence allows organizations to take a proactive approach by identifying threats early and improving their security measures.

Threat intelligence is gathered from various sources, such as security logs, malware analysis, threat feeds, open-source intelligence (OSINT), security vendors, and government agencies.

Benefits of Threat Intelligence

- Helps identify emerging cyber threats.
- Improves threat detection and incident response.
- Supports proactive security monitoring.
- Reduces the risk of successful cyberattacks.
- Assists SOC analysts in making informed decisions during investigations.

2. Describe the Threat Intelligence Lifecycle

The Threat Intelligence Lifecycle consists of several stages that help convert raw data into useful security information.

Stage 1: Planning and Direction

The organization identifies its security goals and determines what information needs to be collected. This stage focuses on understanding the organization's priorities and potential threats.

Stage 2: Data Collection

Relevant information is gathered from different sources such as security logs, firewalls, SIEM tools, threat feeds, malware reports, and public intelligence sources.

Stage 3: Processing

The collected data is cleaned, organized, and converted into a format that can be analyzed. Duplicate or unnecessary information is removed to improve accuracy.

Stage 4: Analysis

Security analysts examine the processed data to identify attack patterns, suspicious indicators, and possible risks. At this stage, the data becomes actionable threat intelligence.

Stage 5: Dissemination

The analyzed intelligence is shared with the appropriate teams, such as SOC analysts, incident responders, or management, so they can take the necessary actions.

Stage 6: Feedback

After the intelligence is used, feedback is collected to evaluate its effectiveness. This helps improve future intelligence collection and analysis.

3. Analyze a Malicious IP Scenario

Scenario

A SOC analyst receives an alert from the organization's SIEM indicating that one of the company's internal computers is repeatedly communicating with an unfamiliar external IP address.

The analyst investigates the alert and checks the IP address against several threat intelligence sources. The IP is listed as malicious and has been associated with ransomware attacks and command-and-control (C2) servers.

The analyst reviews firewall logs and discovers that the infected computer has been communicating with the malicious IP for several hours. Further investigation reveals that the user had opened a phishing email earlier that day, which installed malware on the system.

To prevent further damage, the SOC team immediately isolates the affected computer from the network, blocks the malicious IP address at the firewall, performs a malware scan, and resets the user's credentials. After confirming that the malware has been removed, the system is restored and normal operations resume.

This scenario demonstrates how threat intelligence helps security teams quickly identify malicious infrastructure, investigate incidents, and respond effectively before the attack spreads.